GEMCITABINE

Short infusion (30 min)

Myelosuppression

CHEMICAL NAME 4-amino-1-(2-deoxy-2,2-difluoro-β-D-erythro-pentofuranosyl)pyrimidin-2(1H)-one (hydrochloride)

CLASS Antimetabolite (pyrimidine nucleoside analogue) cytotoxic

TRADE NAMES Gemzar · Gemcitabine injection · Gemita

THERAPEUTIC USES / INDICATIONS

- Pancreatic cancer.
- Non-small cell lung cancer, bladder, biliary, ovarian, breast cancers.
- Lymphoma (salvage).

ADMINISTRATION

- **IV** infusion over 30 min (longer infusion ↑ toxicity).
- Reconstitute with saline; do not refrigerate (crystallisation).
- Cytotoxic precautions.

DOSAGES

- **Pancreatic/lung:** 1,000–1,250 mg/m² on days 1, 8 (and 15) of a 21- or 28-day cycle.
- Dose adjustments based on neutrophils and platelets on treatment day.

PHARMACOKINETICS

- Intracellularly phosphorylated to active di- and triphosphates.
- Deaminated by cytidine deaminase; excreted in urine. **t½:** 42–94 min (intracellular longer).

PHARMACODYNAMICS

- **dFdCTP** incorporates into DNA → masked chain termination; **dFdCDP** inhibits ribonucleotide reductase → ↓ DNA synthesis (S-phase specific).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flu-like symptoms, fever, fatigue.
- Nausea, vomiting (mild).
- Rash, peripheral oedema.

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (dose-limiting).
- Pulmonary toxicity (pneumonitis, ARDS).
- Haemolytic uraemic syndrome; capillary leak syndrome.
- Hepatotoxicity; PRES; radiation recall.

CONTRAINDICATIONS & CAUTIONS

- Severe myelosuppression.
- Pregnancy, breastfeeding.
- Concurrent radiotherapy (severe toxicity) — specialist.

KEY INTERACTIONS

- **Radiotherapy:** radiosensitisation.
- **Live vaccines:** contraindicated.
- **Warfarin:** ↑ INR (reported).

NURSING MANAGEMENT

- FBC before each dose.
- Infuse over 30 min exactly.
- Monitor for breathlessness, oedema, haematuria, hypertension (HUS).
- Paracetamol for flu-like symptoms.

PATIENT EDUCATION

- Flu-like symptoms after treatment are common — paracetamol helps.
- Report breathlessness, swelling, reduced urine, fever ≥38 °C.

KEY POINTS

- Nucleoside analogue; pancreatic and lung cancer.
- Myelosuppression; flu-like symptoms.
- 1,000 mg/m² over 30 min weekly ×2–3.
- Pneumonitis, HUS.

SOURCES CHECKED **[886]** Gemcitabine 10 mg/ml, solution for infusion (UK SmPC) **[887]** Gemcitabine, IV (US label) **[888]** gemcitabine, IV (US label) *Full references p. 432–457*